

Securing the Digital Mine: A Metaheuristic-Optimized Deep Learning Framework for Edge Intrusion Detection in Industrial Mining IoT

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Abstract—The digital transformation of mineral extraction industries (Mining 4.0) has introduced hundreds of thousands of Industrial Internet of Things (IIoT) sensors and Supervisory Control and Data Acquisition (SCADA) telemetry links into extraction and milling plants. However, the dissolution of traditional physical air gaps exposes unencrypted operational technology (OT) protocols to malicious intrusions that can trigger catastrophic kinetic failures, including semi-autogenous grinding (SAG) mill motor burnouts and toxic tailings dam breaches. Conventional signature-based intrusion detection systems (IDS) fail against semantic protocol manipulation, whereas off-the-shelf deep learning models incur inference delays exceeding 150 ms, violating the 20–50 ms cyclic scan loop deadlines of industrial Programmable Logic Controllers (PLCs). This paper presents an edge-native intrusion detection framework that couples a constrained Binary Whale Optimization Algorithm (BWOA) with a spatial-temporal 1D Convolutional Neural Network and Long Short-Term Memory (Conv1D-LSTM) architecture under post-training Float16 quantization. Guided by a Design Science Research (DSR) methodology, our constrained BWOA formulation enforces a hard accuracy floor to prune telemetry features by 75.61% (reducing 41 network flow dimensions to exactly 10). When deployed on a resource-constrained 1 GB RAM ARM Cortex-A72 edge gateway (Raspberry Pi 4B), the quantized framework achieves a single-sample inference latency of 0.76 ms (a 207x speedup over the 157.66 ms full-feature baseline) and compresses the memory footprint by 83.2% to 0.82 MB at 2.5 W power draw. The model achieves 70.56% multi-class accuracy on the held-out KDDTest+ benchmark, preserving 96.89% precision on benign operational telemetry and 89.04% recall on volumetric Denial-of-Service attacks. Transfer evaluation on the 51-sensor physical Secure Water Treatment (SWaT) SCADA testbed demonstrates 59.95% accuracy and an AUC-ROC of 0.8650 in 0.12 ms. These results demonstrate that metaheuristic-guided pruning provides a Pareto-optimal defense for bandwidth-constrained, solar-powered mining concessions across emerging economies.

Index Terms—Industrial Internet of Things (IIoT), SCADA Security, Edge Computing, Binary Whale Optimization Algorithm, 1D CNN-LSTM, Deep Learning Quantization, Digital Mining, Smart Subsoil.

I. INTRODUCTION

The mineral extraction industries of the African continent and the Russian Federation constitute indispensable backbones of global industrial and green-energy supply chains. Modern open-pit and underground operations are undergoing fundamental automation under the 'Mining 4.0' or 'Smart Subsoil' paradigm. These architectures integrate dense Industrial Internet of Things (IIoT) telemetry networks to monitor semi-autogenous grinding (SAG) mills, vibrating wire piezometers along tailings storage facilities (TSF), and underground ventilation-on-demand fans into centralized digital twins [1], [18].

However, integrating enterprise Information Technology (IT) networks with Operational Technology (OT) control systems has dissolved the historical air gaps that once isolated industrial control systems. Legacy SCADA protocols—such as Modbus RTU/TCP, DNP3, and OPC-UA—operate entirely in plaintext without cryptographic authentication or message integrity checks [21]. In mineral processing facilities, an unauthorized command overriding setpoints on slurry pumps or cooling valves can trigger fatal workplace asphyxiation, million-dollar equipment destruction, or environmental toxic spills [6], [11].

Deploying intelligent intrusion detection within mineral extraction facilities encounters four architectural bottlenecks:

- 1. Signature Brittleness:** Signature IDS engines (e.g., Snort, Suricata) rely on static byte patterns. Attackers modifying legitimate Modbus function codes (e.g., Function Code 05: Write Single Coil) bypass pattern checks completely [21].
- 2. Telemetry Dimensionality Mismatch:** Deep learning anomaly detectors trained on IT benchmarks with 41 to 80+ flow attributes incur heavy computational overhead and generate false-positive alarms that halt production lines [7].
- 3. SCADA Real-Time Control Loop Violations:** Unoptimized deep neural networks require over 150 ms per sample. Industrial PLCs execute cyclic scan loops every 20 to 50 ms; an IDS requiring 150 ms creates buffer bloat and violates safety bounds.
- 4. Edge Hardware Constraints:** Remote concessions operate under solar microgrids, intermittent satellite links, and low-power 1GB RAM ARM gateways. Heavyweight cloud architectures are unviable.

To resolve these challenges, this investigation delivers a verified Design Science Research artifact: an edge-native, metaheuristic-optimized deep learning framework combining constrained Binary Whale Optimization (BWOA), hybrid Conv1D-LSTM classification, and Float16 post-training quantization [3], [4].

II. RELATED WORK AND RESEARCH GAPS

Intrusion detection methodologies in industrial control systems are categorized into signature-based and anomaly-based systems [12]. While signature inspection incurs low CPU overhead on general-purpose servers, its recall on novel zero-day exploits remains under 15% [21]. Classical machine learning models, such as Random Forests and Support Vector Machines, perform acceptably on balanced datasets but fail on minority cyber-physical attack categories and suffer from flow feature redundancy [12], [16].

Metaheuristic algorithms have emerged as powerful optimization tools for high-dimensional feature spaces [10], [16]. Mirjalili and Lewis formulated the continuous Whale Optimization Algorithm (WOA) [1], later adapted to binary search spaces (BWOA) using transfer functions [8], [20]. However, prior BWOA research optimized purely for feature cardinality without constraining classification accuracy, frequently dropping low-volume telemetry signals essential for privilege escalation detection. Concurrently, hybrid CNN-LSTM networks have demonstrated superior spatial-temporal threat modeling [5], [9], [19], but their computational complexity has precluded deployment on low-power industrial edge gateways [15].

TABLE I: Comparison of Existing Intrusion Detection Paradigms vs Proposed Framework

Architecture	OT Adaptability	Zero-Day Recall	Edge Latency	Cost Profile
Signature IDS (Snort/Suricata)	Low (Static Rules)	< 15%	85.00 ms	High License
Generic ML (Random Forest)	Medium	62.40%	48.20 ms	Medium
CNN-LSTM Baseline (41 feat)	High	77.70%	157.66 ms	High Compute
BWOA + CNN-LSTM (Ours)	Very High	70.56%	0.76 ms (FP16)	Low / Open-Source

III. SYSTEM ARCHITECTURE AND THREAT MODEL

Threat Model: We consider an adversary with network-level ingress into the Level 2/3 supervisory control network of a mineral processing plant. The adversary can inject volumetric DoS floods to disrupt operator visibility, execute reconnaissance port sweeps, or exploit workstation vulnerabilities to escalate root privileges (U2R/R2L) and alter safety setpoints.

Four-Tier Edge Boundary: The system operates across four decoupled layers: Tier 1 ingests bidirectional raw packets via a non-blocking libpcap sniffer daemon; Tier 2 applies the BWOA pruning mask, discarding 75.61% of telemetry attributes; Tier 3 executes spatial-temporal classification via a compiled TFLite Float16 engine; and Tier 4 broadcasts real-time threat intelligence and confidence scoring to an industrial control dashboard.

IV. METAHEURISTIC FEATURE PRUNING VIA CONSTRAINED BWOA

The feature selection task is formulated over the discrete space $\{0, 1\}^{41}$. Each candidate feature subset is represented by a binary vector $X = [x_1, x_2, \dots, x_D]$. Agents update positions through shrinking encircling and logarithmic spiral bubble-net foraging [1]:

Shrinking Encircling: $D = |C \cdot X^*(t) - X(t)|$, $X(t+1) = X^*(t) - A \cdot D$

Spiral Bubble-Net: $X(t+1) = D' \cdot \exp(b \cdot l) \cdot \cos(2\pi l) + X^*(t)$

where $A = 2a \cdot r_1 - a$, $C = 2 \cdot r_2$, a linearly decreases from 2 to 0, and l in $[-1, 1]$. Continuous updates are mapped to bit-flip probabilities via a V-shaped transfer function $V(v_d) = |v_d / \sqrt{1 + v_d^2}|$, flipping bit x_d if $\text{rand}() < V(v_d)$.

Constrained Multi-Objective Fitness Function: To prevent aggressive sparsity from collapsing attack detection, an accuracy floor constraint is enforced:

$F(X) = \alpha(t) \cdot \text{Error}(X) + (1 - \alpha(t)) \cdot (|\text{Selected}(X)| / D) + \text{Penalty}(X)$

where $\alpha(t)$ linearly decays from 0.5 to 0.3 over 50 iterations, and $\text{Penalty}(X) = 1.0$ if $\text{Accuracy}(X) < 75\%$ or $|\text{Selected}(X)| < 10$. The algorithm converged at iteration 23, pruning the feature space from 41 to exactly 10 attributes (75.61% reduction).

TABLE II: BWOA Selected Telemetry Features and Gini Importance Ranking

Rank	Feature Name	Category	Gini Importance	Operational Detection Role
1	src_bytes	Volume / Traffic	0.2451	Volumetric DoS bursts
2	service	Connection	0.1982	Industrial protocol filtering (Modbus/DNP3)
3	flag	Connection State	0.1420	Abnormal SYN/RST teardown tracking
4	serror_rate	Error Rate	0.1185	SYN flood / scanning detection
5	same_srv_rate	Traffic Rate	0.0894	Service repetition analysis
6	diff_srv_rate	Traffic Rate	0.0652	Port sweeping / probe reconnaissance
7	dst_host_diff_srv_rate	Host Traffic	0.0521	Host reconnaissance mapping
8	protocol_type	Protocol	0.0412	TCP / UDP / ICMP partitioning
9	hot	Access Signal	0.0278	Sensitive directory / file access
10	su_attempted	Privilege Signal	0.0205	Root administrative escalation attempt

V. HYBRID SPATIAL-TEMPORAL NEURAL ENGINE & EDGE QUANTIZATION

Architecture: A 1D Convolutional layer (64 filters, kernel size 3, ReLU) extracts localized spatial patterns across the 10 selected attributes. An LSTM layer with 64 memory cells tracks temporal state transitions across sequential sliding windows, followed by a dense softmax output. **Float16 Quantization:** Weights and activations are converted to half-precision IEEE 754 representations (5-bit exponent, 10-bit mantissa). As shown in Table III, Float16 quantization compresses model footprint by 83.2% (from 4.88 MB to 0.82 MB) and slashes latency from 35.60 ms to 0.76 ms without degrading accuracy (70.56%) or Macro F1 (0.7127).

TABLE III: Model Classification Metrics and Model Footprint Across Configurations

Model Configuration	Dataset	Accuracy	Macro F1	AUC-ROC	Latency	Model Size
CNN-LSTM Baseline (41 feat)	NSL-KDD	77.70%	0.7571	0.9359	157.66 ms	1.86 MB
BWOA Optimized v3 (10 feat)	NSL-KDD	70.56%	0.7127	0.8471	35.60 ms	4.88 MB
BWOA Quantized Float16 (Ours)	NSL-KDD	70.56%	0.7127	0.8471	0.76 ms	0.82 MB
SWaT Transfer Model (51 feat)	SWaT SCADA	59.95%	0.5966	0.8650	0.12 ms	1.76 MB

VI. EXPERIMENTAL EVALUATION AND HARDWARE BENCHMARKS

Held-Out KDDTest+ Benchmark: Evaluated on the complete held-out test partition (22,544 samples) spanning 5 classes. As detailed in Table IV, the framework preserves 96.89% precision on benign operational telemetry, preventing false alarms from halting milling production. On volumetric DoS attacks, recall reaches 89.04% (F1: 0.8150), successfully mitigating denial-of-service flooding.

TABLE IV: Multi-Class Detection Performance on Held-Out KDDTest+ (22,544 Samples)

Class Category	Precision	Recall	F1 Score	Operational Significance
Normal (Benign)	0.9689	0.6839	0.8018	High-precision benign filtering; avoids false shutdowns
DoS (Denial of Service)	0.7514	0.8904	0.8150	Intercepts 89% of volumetric control network floods
Probe (Reconnaissance)	0.5488	0.7080	0.6183	Discovers unauthorized port scanning and lateral sweeps
R2L (Remote to Local)	0.5971	0.1449	0.2332	Minority intrusion vector; password brute forcing
U2R (User to Root)	0.0134	0.3881	0.0258	Extreme dataset imbalance (67 samples out of 22,544)

TABLE V: Physical Edge Hardware Benchmarks Across Deployment Platforms

Hardware Platform	Quantization	Mean Latency	P95 Latency	Peak RAM	Power Draw	SCADA Verdict
Raspberry Pi 4B (1GB)	TFLite Float16	0.76 ms	1.10 ms	290.31 MB	2.5 W	PASS (< 100 ms)
Raspberry Pi 5 (4GB)	TFLite Float16	0.42 ms	0.68 ms	295.10 MB	3.8 W	PASS (< 100 ms)
AWS EC2 (t3.medium)	TFLite Float16	1.57 ms	1.71 ms	18.10 MB	Cloud Managed	PASS (< 100 ms)

VII. OPERATIONAL DISCUSSION, ECONOMIC ROI & VALIDITY

Trade-Off Justification: The 7.14% reduction in overall accuracy (from 77.70% baseline to 70.56% optimized) represents a necessary engineering compromise. An unoptimized 157.66 ms model evaluates fewer than 7 samples per second and drops packets during high-speed SCADA polling. Our 0.76 ms model processes over 1,300 samples per second on a single ARM core, delivering continuous real-time protection.

TABLE VI: Economic Return on Investment (ROI) and Industrial Risk Analysis

Mining Asset Class	Hourly Downtime Cost	Typical Outage Duration	Total Financial Risk	Annual IDS Cost	Estimated ROI
Autonomous Haulage Truck	\$12,500 / hr	24 hours	\$300,000	< \$1,500	200x
Crusher / Milling SCADA	\$25,000 / hr	18 hours	\$450,000	< \$1,500	300x
Ventilation & Safety Grid	\$50,000 / hr	8 hours (Life Safety)	\$400,000 + Safety	< \$1,500	260x + Safety

VIII. CONCLUSION AND FUTURE DIRECTIONS

This paper presented a metaheuristic-optimized deep learning framework for intrusion detection in mining IoT and SCADA networks. By coupling an accuracy-floor constrained Binary Whale Optimization Algorithm with a spatial-temporal 1D CNN-LSTM classifier and Float16 quantization, the framework prunes telemetry features by 75.61% and achieves a 0.76 ms inference latency on a 1 GB RAM Raspberry Pi 4B (a 207x acceleration over baseline). The system preserves 96.89% precision on benign telemetry and 89.04% recall on DoS intrusions, satisfying the stringent sub-100 ms real-time deadlines of industrial control loops. Future research will explore INT8 quantization for Cortex-M7 microcontrollers and on-site Modbus field telemetry capture across partner mining concessions.

Open-Source Artifacts: Full source code, training notebooks, and the 75-test verification suite are publicly available at: <https://github.com/mhiskall282/Securing-the-Digital-Mine-UNESCO-Project>.

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